

ANNEX

DRAFT GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i. e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on

the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted online.
- (ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: [Germany](#)
- Organization(s) or entity(s) responsible for the preparation of the report*: [Federal Foreign Office of Germany](#)
Name: [Mr Harald HERRMANN](#)
Website: www.diplo.de
Email address*:
Phone number*:
Please describe the role/mandate of your organization: [Government Focal Point for the German membership in UNESCO](#)
- Officially designated contact person(s) who completed this survey: [German Commission for UNESCO](#)
Full name*: [Dr Lutz Möller / Dr Fatma Rebeggiani](#)

Position*: Deputy Secretary-General / Programme Specialist Science // Focal Point for the Recommendation for Germany

Email address*:

- Full Name(s) of designated official(s) certifying the report: Mr Harald HERRMANN
- Brief description of the consultation process established for the preparation of the report: The relevant ministries at national and state level have been consulted as well as the ten leading German science organizations (members of "Alliance of Science Organisations": National Academy of Sciences, German Research Foundation, Max Planck Society, Fraunhofer Society, Helmholtz Association, Leibniz Association, DAAD, Alexander von Humboldt Foundation, German Science and Humanities Council, HRK - Conference of rectors of the 300+ German universities, plus the Stifterverband (a joint initiative of German companies and foundations devoted to higher education, science and innovation)).
- Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: none

Website*:

Email address*

Sector: ☐ Public ☐ Private ☐ Civil Society ☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES

If **yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

At federal government level, Federal Ministry for Education and Research (BMBF), as well as Federal Ministry for Digital and Transport (BMDV).

At state government level, the relevant ministries mandated for scientific research of the 16 German *Länder* through the Standing Conference of the Ministers of Education and Cultural Affairs (KMK).

All members of the German "Alliance of Science Organizations" (compare part A).

The 50 most important Learned Societies of Germany.

- 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES

If **yes**, which language(s)?

German (only nationwide official language of the Member State)

- 1.3 Have there been awareness raising activities on open science, including all its key elementsⁱ, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (*ref.: (i) [16.f](#), [16.g](#), [16.h](#)*)

YES

If yes, please provide details on the activities organized or foreseen and links as relevant.

Before focussing on the 2021 Recommendation, the pioneering role of the German science system with regard to open principles in the science system has to be highlighted, since it started already in the year 2003. The “Berlin Declaration” is globally recognized as a key milestone of the Open Science movement; it has created enormous awareness globally and has been signed by some 800 institutions until today. All members of the German “Alliance of Science Organizations” have signed the Berlin Declaration. The OA2020 initiative has later added new momentum to the Berlin Declaration.

As concerns the 2021 Recommendation, the German Commission for UNESCO has organized about fifteen conferences and workshops to raise awareness about the Recommendation, starting in 2020 with the start of the negotiations. These activities targeted different audiences in Germany and internationally. As invited speakers to other conferences, representatives of the German Commission for UNESCO contributed to at least another fifteen Open Science events in Germany.

Several key Open Science stakeholders of German science organizations have included the 2021 Recommendation prominently into their standard presentations and their websites. Major funders and science organizations refer to the recommendations in their own Open Science Positions or Policies, see below 1.4.

In March 2022, the international Open Science Conference 2022 hosted by the Leibniz Association (with several hundred participants) was co-organized with the German Commission for UNESCO. Its main panel discussed the 2021 Recommendation, with participation (inter alia) from UNESCO Secretariat and the current Chair of the UNESCO Executive Board. In addition, a diverse culture of Open Science event formats has developed in Germany over the years (e.g. Open Science Festivals, thematic conferences and meetings).

In 2024, the German Commission for UNESCO has published a brochure with the translation of the Recommendation into German alongside the English original.

In 2024, the German Commission for UNESCO hosted a global online seminar for National Commissions to prepare this very 2025 periodic reporting procedure.

In October 2024, the European Open Science Cloud (EOSC) Symposium was held in Berlin, attracting more than 600 participants from Germany, Europe and abroad.

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (*ref.: (i) [16.a](#), [16.b](#), [16.c](#), [16.d](#), [16.e](#), [16.h](#)*)

YES

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

In 2022, the Helmholtz Association of large German research facilities adopted its new Open Science Policy. In its very first line, the Policy refers to the 2021 Recommendation as basis of the Helmholtz Association's commitment to Open Science (<https://os.helmholtz.de/en/open-science-in-helmholtz/open-science-policy/>).

In 2022, the Leibniz Association of 96 research institutes that address issues of historical, social, economic, technological, health, and ecological relevance adopted its new Open Science Policy. The Policy uses almost all definitions in an identical way to the 2021 Recommendation and also refers to UNESCO explicitly (<https://zenodo.org/records/13777926>).

In 2022, the German Research Foundation (DFG) published its position paper on "Open Science as part of a Culture of Science". On its first page, the document refers to the 2021 Recommendation "as an essential basis for the further development of open science" (<https://www.dfg.de/en/news/news-topics/announcements-proposals/2022/info-wissenschaft-22-79>).

The joint focus "Digitalisation in Science" of the German "Alliance of Science Organizations", established already in the year 2008 as the joint initiative "Digital information", ensures consistent action of the German science system. While the initial focus was on improving access to digitally available information, the focus has been significantly widened to cover most Open Science principles in line with the UNESCO 2021 Recommendation in recent years.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?

YES

If yes

- please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

Before answering this and the following questions specifically for Germany, it has to be highlighted that the STI policies of the European Union and its associated instruments, entities and funding programmes, in particular Horizon Europe and the European Research Council, are very important for the German STI system. The European Open Science Cloud (EOSC) with its recently (October 2024) launched service platform EOSC EU Node will likely be the most important Open Science platform in Europe, and therefore for Germany. Another recent example is the European Commission's Revised Charter for Access to Research Infrastructures to foster open science, innovation, and research security. On a national level, the development of the European Research Area is supported via the German ERA Roadmap by the Federal Ministry for Education and Research (BMBF).

However, assuming that there will be a separate coverage of the European level, all following answers to all questions will focus specifically on the German level.

Document and details: "Zukunftsstrategie" or "Future Research and Innovation Strategy"; Federal Ministry for Education and Research (BMBF); developed in cooperation with advisory bodies of the ministry such as the Experts' Commission EFI, the Hightech Forum, the Innovation Dialogue and the OECD

Review of Innovation Policy Germany
https://www.bmbf.de/SharedDocs/Publikationen/DE/1/730650_Zukunftsstrategie_Forschung_und_Innovation.pdf?__blob=publicationFile&v=4);
 predecessor: Hightech Strategy in 2006, 2010, 2014 and 2018; 2023; English
 version of the Executive Summary:
www.bmbf.de/SharedDocs/Publikationen/DE/FS/747580_Zukunftsstrategie_Forschung_und_Innovation_en.pdf)

- does it include any of the following open science elements:
 - open access to scientific knowledge YES, page 66/67 of full version
 - open science infrastructure YES, page 66/67 of full version
 - open engagement of societal actors YES, page 81/82 of full version
 - open dialogue with other knowledge systems YES, page 24 of full version

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h*)

YES

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

Secondary Publication Right; German Copyright Law; 2014; covers scholarly publications which are publically funded; German version:

https://www.gesetze-im-internet.de/urhg/_38.html

Open Data Strategy; German Federal government; 2023; covers specifically also research data; English version:

<https://bmdv.bund.de/SharedDocs/DE/Anlage/K/driving-progress-with-data.pdf>

Open Access in Germany, Joint Guidelines; Federal Government and the Länder; 2023; English version:

http://www.bmbf.de/SharedDocs/Publikationen/DE/FS/772970_Open_Access_in_Deutschland_en.pdf

National Strategy on Open Educational Resources; Federal Ministry for Education and Research (BMBF); 2023; German version (translation currently not available)

www.bmbf.de/SharedDocs/Publikationen/DE/3/691288_OER-Strategie.pdf

White Paper Citizen Science Strategy 2030 for Germany; consortium of several hundred representatives of German science; 2022; English version:
<http://zenodo.org/record/7117771>

Adopting a Research Data Law at national level is currently under discussion.

Recommendations on the Transformation of Academic Publishing: Towards Open Access; German Science and Humanities Council, 2022: https://www.wissenschaftsrat.de/download/2022/9477-22_en

Federated Data Infrastructures for Scientific Use. NFDI, EOSC, Gaia-X, and the European Data Spaces: Comparison and Recommendations for a Committed Engagement to Shape the European Research Data Ecosystem; Rat für Informationsinfrastrukturen; 2024: <https://rfii.de/download/federated-data-infrastructures-for-scientific-use/>

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.~~

2.3 In your country, are there specific policy instruments^v that aim to promote open science?

YES

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included*.

There are several such specific policy instruments already in existence and more are under development. It is impossible to list them all. The following short selection focuses on some of the most important instruments.

The National Research Data Infrastructure (Nationale Forschungsdateninfrastruktur, NFDI) has been founded in 2019, upon initiative of the Council for Information Infrastructures (RfII). The goal of NFDI is to establish a nation-wide, decentral and ever-growing network of infrastructure and services for research data management according to agreed standards and the FAIR principles. One key goal is to foster the re-use of research data in and across all disciplines. NFDI has been established as a private-law organization and has currently more than 300 members. 27 NFDI consortia in different scientific disciplines are funded (with up to 90 million Euro per year until 2028) by the national and the state governments. The NFDI is considered as the main German (in-kind) contribution to the European Open Science Cloud (EOSC). With respect to EOSC, NFDI is among the potential candidates for the enrolment of the first wave of Candidate EOSC Nodes and currently part of the first-stage dialogue meetings (<https://eosc.eu/news/2024/11/eosc-tripartite-governance-establishes-process-to-sequence-enrolment-of-candidate-eosc-nodes/>).

GO FAIR is an international initiative with focus on Europe that has been jointly created by France, Germany and the Netherlands, with a shared funding of all three governments and a shared coordination office by the three countries. GO FAIR has as key purpose to promote the FAIR principles in the context of the EOSC.

Open Access is the element of Open Science with the largest diversity of policy instruments in Germany:

The DEAL Consortium is initiated by the Alliance of Science Organizations in Germany, spearheaded by the German Rectors' Conference (HRK) and operated by the Max Planck Digital Library (MPDL). DEAL negotiates nationwide transformative and ground-breaking "Publish and Read" agreements with the largest three commercial publishers of scholarly journals on behalf of the German research institutions landscape. All scientific publications published under DEAL's framework contract rules are accessible Open Access. By participating in the DEAL agreements, German scientific institutions receive a uniform, comprehensive service package that covers access to the entire range of these journals. These access rights are also permanently secured by DEAL. Currently, DEAL started an "Open Access means CC-BY" initiative. The most recent five-year "opt-in" contract with Elsevier was signed in 2023, the second with Springer Nature in 2023, and also the second with Wiley in 2023, enabling approximately half of German research outputs to be published Open Access. The aim is to enable researchers in Germany to publish openly. At the same time, it aims to contain the costs of scientific publishing and establish a sustainable, publication-based funding model. (<https://deal-konsortium.de/en/about-deal/rationale-and-objectives>).

The German Research Foundation's (Deutsche Forschungsgemeinschaft) funding programme 'Open Access Publication Funding' is of great significance to promote Open Science. The Programme grants a fixed allowance for the publication of research results via Open Access to research institutions (not individuals). In 2024, grants worth more than 38 million Euro have been awarded (funding rate of 93%). To foster equity along the lines of the UNESCO recommendations, Diamond Open Access and publically funded infrastructures for Open Access are supported as well, e.g. the National Service Center for Diamond Open Access (<https://www.dfg.de/en/news/news-topics/announcements-proposals/2025/ifr-25-07>). Further funding programmes target establishing and developing infrastructures for research data management and – since 2024 – for research software also.

The Max Planck Society has a significant impact both nationally and globally through its dedication to advancing Open Access scholarly publishing. Beginning in 2003, as initiator of the Berlin OA Conference series, the Max Planck Society has continued to provide one of the most important forums for professional discussion of the international development towards Open Access publishing of scientific knowledge. Over the years, this conference series has evolved and is now organized by Open Access 2020 (OA2020), a dedicated initiative that continues to convene international stakeholders to drive Open Access forward. OA2020, coordinated by the Max Planck Digital Library (MPDL), brings together research institutions and funders worldwide to coordinate and enhance negotiation power with publishers and to foster policy development. At the national level, the Max Planck Society operates the MPDL Services gGmbH, which was established to implement nationwide Open Access transformation agreements under the DEAL project (see above).

The German Ministry for Research and Education is funding the establishment of an Open Access culture as central part of Germany's scientific landscape in diverse funding programmes and initiatives, ranging from the national open-access.network competence center for Open

Access to concrete projects for establishing new Open Access formats and business models.

As concerns Open Science in truly all its facets, activities of the Leibniz Association and of the Helmholtz Association stand out. The Leibniz Strategy Forum on Open Science (of the Leibniz Association; the name of the format was previously the Leibniz Research Alliance Open Science) foster Open Science practices within the Leibniz Association and beyond. For more than 10 years, it hosts the annual international Open Science conferences. In turn, the Helmholtz Open Science Office has been founded in 2005. It has established an Open Science task force across all Helmholtz Association research facilities, it provides consultancy and advice, published practical recommendations, guidelines and briefings and hosts diverse meetings (e.g. more than 70 online seminars).

Also non-governmental international initiatives have to be highlighted that have impact in Germany, such as the "Amsterdam Declaration on Funding Research Software Sustainability", which has been signed in 2024 by the DFG, for example. Similarly, the 2024 "Barcelona Declaration on Open Research Information" has been signed by several higher education institutions and academic libraries from Germany.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.~~

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (*ref.: (ii) [17.a](#), [17.b](#), [17.c](#), [17.f](#), [17.h](#)*)

NO

~~If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.~~

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.~~

In light of existing strategies for Open Data, Open Access, OER, Citizen Science (cp. above), the need for such an instrument was not seen so far. Discussion is ongoing.

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (*ref.: (ii) [17.d](#), [17.e](#), [17.g](#)*)

YES

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and [elements of open science](#) included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

Compare question 1.4 as refers to research associations and organisations. Additionally, many if not most higher education institutions of all types and sizes in Germany have implemented Open Science strategies and/or policies. There is no complete overview on the national level due to the

federal structure of the German research and education system. Some pertinent examples: Berlin University Alliance (HU Berlin, FU Berlin, TU Berlin, and Charité Berlin), University of Freiburg, University of Tübingen, Leibniz University Hannover, TH Köln, the universities in Hamburg, University of Applied Sciences Jena.

- 2.6 In your country, are there specific funding mechanisms^{vi} for open science at the national level? (*ref.: (ii) 17.j. (iii) 18.j. (iv) 19.c*)

YES

If yes or under development, please provide details and links as relevant.

~~If no~~, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

Compare question 2.3

- 2.7 In your country, have open science practices^{vii} been integrated in existing research funding mechanisms?

YES

If yes, please provide details and links as relevant.

German research funders have not made Open Access practices a legally obligatory requirement for recipients of research funding in general. However, a change of German copyright law made a secondary publication according to Open Access principles a legally possible choice for the authors in the first place (compare the act UrhG).

Also at the level of the *Länder*, there exist funding programmes for Open Science practices. For example, the “NiedersachsenOPEN” programme, with a total fund of 3.5 millions EUR, supports non-commercial, science-based publication programmes based on ‘Diamond Open Access’ and, beyond that, the development of a ‘Diamond Open Access publication landscape’ in the state of Lower Saxony (www.mwk.niedersachsen.de/startseite/aktuelles/presseinformationen/niedersachsen-fordert-open-access-publikationen-mit-neuem-fonds-230331.html). Similar funds to support Open Access publications exist in Schleswig-Holstein and Thuringia.

Researchers employed by certain types of institutes, for example those of the Helmholtz Association, are required to publish Open Access (if there are no contravening legal requirements).

The Alexander von Humboldt Foundation also supports the Open Access publication of research results.

There are ongoing discussions in the German Alliance of Science Organizations and in the German National Chapter of CoARA members as to whether and how to include Open Science practices in Research Assessment. The current view is that while Open Science activities should, where appropriate, be recognized as part of the assessment of individual researchers, it is unwise to treat (the degree of) Open Science implementation as a criterion in the assessment of research proposals, as there are currently more effective ways to support Open Science.

~~If no~~, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

YES

If yes, please provide details and links as relevant.

For the DEAL consortium, please refer to question 2.3. The Stifterverband has promoted the adoption of Open Science strategies and policies by higher education institutions in different ways. E.g. it has developed an Open Science self-assessment tool for universities (<https://road2openness.de/en/>) as well as set up an Open Science strategy workshop programme to develop such a strategy with workshops, consultation hours, dialogue and networking. For the call for applications see www.stifterverband.org/road2openness-strategiewerkstatt.

Specifically with regard to research data, there are intense discussions how partners from industry and publicly-funded research can exchange their data and enter into collaboration agreements. In this regard, the NFDI section "industry engagement" (<https://www.nfdi.de/section-industry-engagement/?lang=en>) is an important player.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.i](#), [23.c](#))

Not on a national level. However, there is German involvement in the OSMI, the Open Science Monitoring Initiative (<https://open-science-monitoring.org/>), with the Charité Berlin as a founding member.

~~If yes or under development, please provide details and links as relevant.~~

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

In 2023, Germany invested 3,1% of its GDP in research and development (<https://www.stifterverband.org/forschung-und-entwicklung>; <https://www.destatis.de/Europa/DE/Staat/EU-Staaten/Deutschland.html>).

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

In Germany, almost the entire population has access to reliable internet with sufficient bandwidth. In 2021, 98.3% of urban households and 82,3% of rural households had access to broadband of at least 50 Mbit/s. Mobile internet

comes on top (in 2024, 99.8% coverage of 2G and 92.5% of 5G) (BMVI 2021: https://bmdv.bund.de/SharedDocs/DE/Publikationen/DG/breitband-verfuegbarkeit-mitte-2021.pdf?__blob=publicationFile)

- 3.3 Are there national research and education networks (NRENs)^{viii} active in the country? (*ref.: (iii) 18.c*)

YES

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

The “German Research Network” (DFN) was established in 1984. It connects some 850 universities and other research institutes to the X-WiN (most recent network infrastructure with 10,250 kilometers of optical fiber connections and a multi-terabit core network) and Géant. X-WiN is one of the largest and most powerful communication networks in the world. www.dfn.de/en/network/

~~If no~~, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

- 3.4 Are there national or institutional [open science infrastructures](#)^{ix} in your country? (*ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m*)

YES

If yes or under development, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- platforms for exchanges and co-creation of knowledge between scientists and society
- community-based monitoring and information systems to complement national, regional and global data and information systems.

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

All infrastructure suggested in this question is available in a non-commercial way to German public sector scientists. Many or most infrastructures respect the FAIR principles. Through the work of NFDI consortia, the CARE principles get more and more known. Many infrastructures are offered at the level of the 16 *Länder*, because the *Länder* have the constitutional mandate for education and higher education. Providing a list of all such infrastructures is therefore impossible because of the high number. However, there is a number of relevant registries for publication repositories (Liste der Publikationsdienste - Deutsche Initiative für Netzwerkinformation e. V., <https://dini.de/dienste-projekte/publikationsdienste>), research data

repositories (<https://www.re3data.org/>) and other reliable and certified infrastructures (https://resources.dfg.de/home_en.html).

The UNESCO Open Science Outlook 2023 contained the information from May 2023 that Germany with 498 Open Data repositories ranked No 2 globally and with 299 Open Access repositories ranked No 4 globally. In addition, there are about 345 Diamond Open Access journals.

According to a study by the Technopolis Group (2023) commissioned by the German Research Foundation for the Alliance of Science Organizations to map and describe Open Access services in Germany, non-commercial Open Access offerings in Germany form an important (and importantly decentralised) backbone of the academic publishing culture, which includes sufficient subject-specific and interdisciplinary publication offerings and background services. However, there remains a lack of success criteria as well as coordination for avoiding redundancies. In addition, not all services are yet reliable, sustainable, and competitive in terms of their quality.

For the nation-wide developed standards for research data management in the NFDI project, compare question 2.3.

Germany hosts several high-performance computers at public research facilities. The three computers that are currently offering more than 5 PetaFLOPS in peak capacity are HAWK in Stuttgart (26 PetaFLOPS), LEVANTE at the German Climate Research Center Hamburg (14 PetaFLOPS) as well as JUWELS at Jülich Research Center (85 PetaFLOPS). In 2025, with JUPITER, the first European ExaFLOPS computer will be launched in Germany.

Furthermore, the NHR Alliance (<https://www.nhr-verein.de/en>) bundles the resources and competencies of university high-performance computing and make them available to scientists at German universities free of charge. The NHR Alliance does not only provide computing capacities, but also supports users e.g. by training in the use of high-performance computing in their fields of application.

Most of the 16 *Länder* offer a state-wide academic cloud to their universities (e.g. Academic Cloud in Lower Saxony: <https://academiccloud.de/>).

One of the most important infrastructures for Open Science are “support offices” that provide advice and consultancy at the level of some of the 16 *Länder*. Examples are:

- the openaccess.nrw network and collaboration of the state of Northrhine Westfalia
- “Hamburg Open Science” of the state of Hamburg
- the Open Access office of the state of Berlin

Such state-based infrastructures are often based on state Open Science strategies

- the E-Science strategy of the state of Baden-Württemberg
- the Open Access strategy of the state of Brandenburg
- the state of Schleswig-Holstein’s strategy 2020
- the state of Thuringia’s strategy for a digital society

In the context of Open Access, these are anchored on a national level by BMBF's Open Access strategy from 2016, the Bund-Länder-Open-Access-Guidelines from 2023 and BMBF's establishment of nation-wide infrastructure like the open-access.network competence platform and the Open Access Monitor.

As concerns Citizen Science, the networking platform www.mitforschen.com (earlier: www.buergerschaffenwissen.de) is the central national platform for Citizen Science in Germany. The platform presents, connects and supports citizen science projects since 2014 and is funded by the BMBF.

Since the year 2000, the BMBF has funded the initiative "Science in Dialogue" ("Wissenschaft im Dialog", WiD, <https://wissenschaft-im-dialog.de/>) as the most important nationwide structure to foster science communication. WiD offers the annual national forum on science communication. There is also NaWiK (<https://nawik.de/>), the national institute on science communication, and the national association of higher education communication (<https://www.bundesverband-hochschulkommunikation.de/>). All three partners provide together the platform www.wissenschaftskommunikation.de and together or individually they offer several additional nationwide formats such as training, newsletters, etc. All formats receive funding by the BMBF.

Currently, a "National Service Point for Diamond Open Access" is being established in Germany (compare question 2.3) Its aim is to promote the establishment and further development of diamond open access publication structures. As the German 'Diamond Capacity Centre', the Service Agency will be part of a European network in the European Diamond Capacity Hub (EDCH).

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: [\(iii\) 18.e, 18.f](#))

YES

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open scienceⁱ, stage of research, accessibility to all the relevant users in the region or internationally).

Germany is a key supporter of the European Open Science Cloud (EOSC). The EOSC Association based in Brussels has 19 institutional members from Germany, among them NFDI (compare question 2.3), the DFG, the DFN (cp. question 3.3) and individual institutions such as DESY, DLR and GSI. The NFDI is the officially mandated organisation in the EOSC association and is considered at the main (in-kind) contribution of Germany to the EOSC. In addition, a process is currently underway to select so-called national EOSC nodes that complement the EOSC EU node that was presented to the public in October 2024. The EOSC EU node currently offers six services open for European researchers, among which file sync and share, virtual machines, large file or bulk data transfer and a cloud container platform.

Furthermore, Germany supports the expansion of Open Research Europe as a publicly funded publication infrastructure.

It also funds or contributes to funding international infrastructures like zbMath, SCOAP3, DOAJ and OAPEN.

~~If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?~~

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: [\(iii\) 18.g](#))

YES

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open scienceⁱ, stage of research, accessibility to all the relevant users in the region or internationally).

Germany supports North-South and North-South-South collaborations in research with all world regions. Moving the frameworks of such collaborations towards equity among participants in the collaborations is a priority. At the same time, Open Source principles are also actively promoted in such collaborations, sometimes with considerable funding. This is especially true in collaborations that aim at establishing long-term viable research infrastructures. However, it has become more and more important to take security and competition aspects into account when implementing Open Science principles worldwide.

An example for this support is SASSCAL (Southern African Science Service Centre for Climate Change and Adaptive Land Management), established in 2012 with funding from BMBF. Part of SASSCAL is the transnational OADC (Open Access Data Center, <https://sasscal.org/oadc-open-access-data-center/>). For West Africa, WASCAL (West African Science Service Centre on Climate Change and Adapted Land Use), also established in 2012, with funding from BMBF, is working on an open data infrastructure.

Open Science principles are also actively promoted in smaller-scale projects such as in the 2021-2023 project OWASA (Open science to support local water security in Southern Africa) as part of the BMBF funded African-German research programme Water Security in Africa - WASA (<https://www.fona.de/en/measures/funding-measures/water-security-in-africa-wasa.php>), providing observation and modelling data to government stakeholders and research institutions in Africa.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a, 19.c](#))

YES

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

Building the "[open-access.network](#)" started in 2019 and is now, after completion, a nation-wide network for capacity building on Open Access,

funded by BMBF. The network offers full information on Open Access, and clarifies any question and presents benefits and challenges. The network aims at connecting relevant Open Access pioneers and existing Open Access service points in higher education institutions and research performing institutions and offers various trainings, workshops and webinars. A large variety of Open Educational Resources and other Open Access competence building material is offered by open-access.network to be included in competence building measures at higher education or research institutions.

There are similar platforms for research data (<https://forschungsdaten.info/>) and for Open Educational Resources (<https://open-educational-resources.de/>).

Universities in Germany are autonomous. Therefore, each of the 400+ German universities decides individually about the capacity building it offers to its students, early-career scientists and seasoned scientists. Many or most universities offer regular formalized courses on Open Science. Often, such courses are provided by the university library. To our knowledge, there are no obligatory courses for students yet at German universities.

As an example, since 2023, the University of Konstanz which is a leading German university in terms of Open Science, offers (in English language and as OER) a basic course and an advanced course on Open Science, with altogether 15 online modules of 15 minutes each: www.kim.uni-konstanz.de/en/openscience/open-science-courses/. As another example, the University of Bochum has hosted its first Open Science Month with a Moodle course and more than 20 events in November 2024, after hosting Open Science weeks in previous years. Another pioneer is Munich's LMU university which in 2018 has founded the LMU Open Science Center (<https://www.osc.uni-muenchen.de/>).

German Science Organisations do also engage in capacity building. The Max Planck Society e.g. is active through regular initiatives aimed at researchers. A key component of these efforts are the periodically organized Open Science Days, a conference that hosts open expert discussions on current developments within Open Science within and beyond the Max Planck Society. This is complemented by the annual meeting of the Open Science Ambassadors, a network primarily composed of early-career researchers who serve as multipliers within their Max Planck Institutes. The meeting includes dedicated talks, workshops and discussion on potential activities to further promote Open Science practices across the Max Planck Institutes.

In addition, the consortia of the National Research Data Infrastructure (NFDI; cf. 2.3) offer a wide range of discipline-specific training workshops, tools or help desks to support researchers. German institutions also actively contribute to the annual international Love Data Week (LDW), an international week of action around the topics of research data and research data management, with a very broad and varied programme (<https://forschungsdaten.info/fdm-im-deutschsprachigen-raum/love-data-week-2025/>).

In 2021 and 2022, the BMBF implemented three calls for proposals on data competencies in research: one focussing on young researchers, one on data management in universities of applied sciences and one with a structural component with the aim to build-up 11 data competence centres (Datenkompetenzzentren) at German higher education and research institutions. The projects are currently running.

The *Länder* support the Kooperationsnetzwerk OER-förderliche Infrastrukturen und Dienste (cooperation network OER-promoting infrastructures and services; KNOER). KNOER is an association of mandated public institutions that promote, coordinate and network digitally supported teaching and learning in their respective higher education institutions.

With the KNOER cooperation network, the participating institutions have created a binding and sustainable structure for cross-Länder cooperation in the digitisation of teaching in the higher education sector. The cooperation is geared towards cross-Länder networking and bundling of OER-conducive, digital teaching/learning infrastructures and services. The network enables its members to exchange experiences as equal partners, identify common interests and represent them at the federal level. In doing so, they jointly advance important issues and solutions around OER in higher education.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES

If yes, please provide more information and links as relevant.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

Germany has a favourable political framework for Open Science and is making good progress in this respect. It is the subject of numerous ongoing initiatives and open science activities and is partly covered by funding activities (e.g. BMBF-funded programme on data competence centres).

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (*ref.: (iv) 19.b*)

NO

~~If yes, please provide more information and links as relevant.~~

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Open Science (or relevant parts of Open Science such as science communication or Open Access) are part of many individual higher education curricula. Due to university autonomy, only very few centralized requirements are possible for higher education curricula in Germany. Higher education curricula in general are not approved by state authorities but subject to quality assurance procedures based on a peer process run by private-sector accreditation agencies and the German Accreditation Council. Therefore, it is impossible for the Member State to require the incorporation of such competencies in curricula. Instead, it is incumbent upon the Member State to highlight the importance of Open Science practices and count on voluntary inclusion into curricula. This is happening.

- 4.4 Are open educational resources^x as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

YES (compare question 4.1)

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))

YES

If yes, please provide more information and links as relevant.

Compare the list of initiatives in 3.4

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b, 20c, 20i](#))

YES

If yes, please provide more information and links as relevant.

German science organizations participate in several international initiatives to review current practices of research assessment. For example, CoARA has more than 30 members from Germany, including the DFG, the Leibniz Association, the Alexander von Humboldt Foundation and several large universities. A National Chapter for CoARA was created in 2024 (<https://www.dfg.de/de/aktuelles/neuigkeiten-themen/info-wissenschaft/2024/ifw-24-17>). The EU Commission has established in 2024 an action plan to implement CoARA recommendations (ARRA).

The Global Young Academy, whose Secretariat is hosted in Germany, is also involved in several initiatives.

German researchers also provided much input to the debate on quality criteria for societal engagement in research.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.c, 20.d](#))

NO

~~If yes, please provide details and links as relevant.~~

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There are early debates on this issue and the Alliance of Science Organizations held a workshop on this topic in 2024 preparing a document to be made public in 2025. The German Research Foundation published a thorough analysis of currently obstacles in Research Assessment as relating to publications: DFG, German Research Foundation - Academic Publishing as a Foundation and Area of Leverage for Research Assessment (<https://www.dfg.de/resource/blob/175776/d2d371e8572b7bcf23bd8bf273354ac6/positionspapier-publikationswesen-en-data.pdf>)

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h](#))

YES

If yes, for each initiative, please provide more information, including on the stakeholders^{xi} that are involved or lead the initiative, and links as relevant.

There are (and have been) several awards for Open Science practices. For example, the “Enter Award” for Open Access practices, funded by the BMBF and implemented by [irights.lab](#) in 2024. The Hamburg Open Science Programme, the Humboldt University of Berlin, the University of Lübeck, the University of Oldenburg, and the Max Planck Institute for Educational Research have also established such awards. Also, the Berlin Institut of Health @ Charité offers a range of awards which aim at diverse dimensions of open science: <https://www.bihealth.org/en/translation/innovation-enabler/quest-center/calls-and-awards>. The Stifterverband is one of the key players to support Open Science practices and as such offers e.g. the Open Data Impact Award (<https://www.stifterverband.org/innosci/open-data-impact-award>).

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 5.4 Have there been any unintended negative consequences^{xii} of open science practices reported in your country? (ref.: [\(v\) 20](#))

YES

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

There is an intensive debate in the German science system on how to promote openness in international cooperation with partner countries with other practices and understanding of openness. Negative consequences are currently also discussed in relation to the costs for (commercial) Open Access. Open Science has become a topic in relation to Research Security and Digital Sovereignty.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (*ref.: (vi) 21*)

~~YES, at the national level.~~

~~YES, at the institutional level.~~

YES, at both national and institutional levels.

NO

If **yes**, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
- promoting open innovation practices.
- Others (with a box to add text)

All these practices are promoted or at least discussed and experimented with at national and at institutional level. Most of these issues have been addressed before in this questionnaire. In particular, there are examples of higher education institutions' Open Science policies and strategies that address several of these actions (see above). Also, the NFDI is an important actor contributing to sharing data openly in and across disciplines.

Many of these aspects are covered by the "[Code of Conduct: Guidelines for Safeguarding Good Research Practice](#)" (administered by the DFG) that is adopted by almost all universities and research organizations. Regarding pre-prints and open review as well as sharing negative results, see e.g. the DFG report "[Academic Publishing as a Foundation and Area of Leverage for Research Assessment](#)" (cf. 5.2). As concerns Open Innovation as a part of Open Science, Germany has supported the 2024 consensus in the G20 on an Open Innovation Strategy. Also as concerns Open Innovation, the "Stifterverband" of German foundations has implemented a multi-annual project in recent years (innOsci) that has promoted the strategic opening of innovation processes. The project has resulted, inter alia, in an "Open Transfer Index" to measure open innovation processes. Two examples with considerable use in Germany can be given as concerns participatory strategies for highlighting socially relevant issues: "Bürgerräte" (bringing together dozens or hundreds of randomly chosen citizens), inter alia the "Bürgerrat Forschung" (citizens' council on research), as well as "Reallabore" ("living labs") as structured cooperation formats between science and civil society. Both formats are used at the local level and at the national level.

With regards to Citizen Science, the Leibniz Association promotes the concept through its internal working group Citizen Science, dozens of concrete projects and participation in the European Citizen Science Association. The Helmholtz Association pursues exactly the same approach. The Alliance of Science Organizations in Germany has issued a statement in 2022 where it commits itself to enable participation in research and to support it where it promises added value for science and the humanities and society. It also defines certain premises for fruitful citizen participation in science (https://www.allianz-der-wissenschaftsorganisationen.de/wp-content/uploads/2022/11/2022-11-09_Allianz_Statement_Participation.pdf)

See also the BMBF funded internet portal «mit:forschen» - <https://www.mitforschen.org> . The 6th Forum Citizen Science was held in 2023 in Freiburg.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

- 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a\)](#))

Relevant strategies of German international scientific cooperation, for example the 2024 Strategy for promoting an international orientation of German universities (https://www.kmk.org/fileadmin/veroeffentlichungen_beschluesse/2024/2024_06_14-Strategie-Internationalisierung-Hochschulen.pdf), contain implicit or explicit references to Open Science principles – already the predecessor 2016 strategy promoted North-South collaborations on Open Access.

- 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b, 22.e, 22.f\)](#))

NO

~~If yes, please provide links as relevant and more information.~~

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

It is a priority to promote Open Science principles as part of all international activities. A stand-alone plan on cross-border Open Science practices is not a priority thus.

However, Germany is an active member of CoNOSC, the Council for National Open Science Coordination. CoNOSC brings together national Open Science leaders to engage in an international dialogue and to share good practice for a stronger and more unified, workable Open Science policy framework (<https://conosc.org/>).

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: [\(vii\) 22.c](#))

YES

If yes, please provide links as relevant and more information.

European Open Science Cloud (<https://eosc.eu/> and <https://open-science-cloud.ec.europa.eu/>) – INFRAEOSC

Open Research Europe (Statement of Intent on Open Research Europe : <https://zenodo.org/records/14624287>)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

NO

If yes, please provide more information.

-
- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
- ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
- iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
- iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
- v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
- vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
- vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
- viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
- ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories,

software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))

xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.